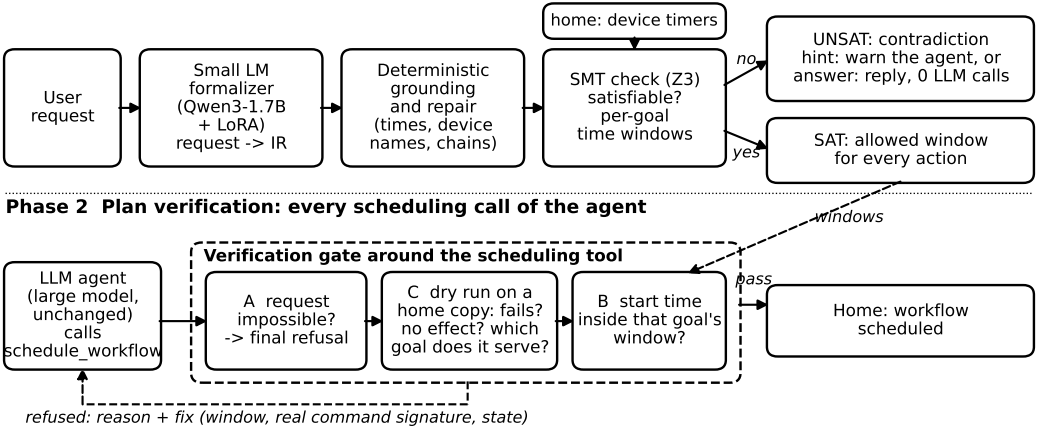
Phase 1 Pre-flight: once per request, no large-model call
User request
Small LM formalizer (Qwen3-1.7B + LoRA) request -> IR
Deterministic grounding and repair (times, device names, chains)
home: device timers
SMT check (Z3) satisfiable? per-goal time windows
no
yes
UNSAT: contradiction hint: warn the agent, or answer: reply, 0 LLM calls
SAT: allowed window for every action
Phase 2 Plan verification: every scheduling call of the agent
windows
Verification gate around the scheduling tool
LLM agent (large model, unchanged) calls schedule_workflow
A request impossible? -> final refusal
C dry run on a home copy: fails? no effect? which goal does it serve?
B start time inside that goal's window?
pass
Home: workflow scheduled
refused: reason + fix (window, real command signature, state)